DIGGIT

Digital Interface for Graphing Geochemical Interpolation Treatment

PLATFORM OBJECTIVE — DEVELOPMENT FRAMEWORK

Mission statement
To develop a collaborative environmental research application integrating satellite-derived observations, geospatial datasets, site-specific environmental measurements, scientific literature and predictive analytics into an interactive GIS interface. DIGGIT will use AI-assisted research, geochemical interpolation and climate-aware forecasting to evaluate remediation options, support evidence-based protocol selection, and coordinate remediation schedules while minimizing financial costs, environmental disturbance and implementation delays.

1. The integrated data environment
The platform should combine four principal data domains.

A. Satellite and geospatial intelligence

- GSAT data source (acronym/source to confirm)
- Copernicus Sentinel-2
- Landsat
- Terrain, land cover, vegetation indices, surface temperature where available, hydrology and historical change.

B. Site-specific measurements

- Soil and sediment chemistry.
- Groundwater and surface-water measurements.
- Soil moisture, pH, redox potential and conductivity.
- Contaminant concentrations, depth profiles and sample locations.
- Microbial, fungal and vegetation observations.

C. Scientific research and analytical evidence

- Peer-reviewed remediation studies.
- Organism–contaminant compatibility evidence.
- Degradation pathways and transformation products.
- Sorption, bioavailability and transport characteristics.
- Laboratory, pilot and field-validation results.

D. Climate and operational intelligence

- Rainfall, temperature, evaporation and drought.
- Seasonal groundwater and soil-moisture variation.
- Flooding, erosion and runoff risks.
- Treatment windows and weather-sensitive operations.
- Budget, resource and scheduling constraints.

2. The geospatial GUI
The map should be the primary workspace, not an accessory to a collection of charts.

I envision a central interactive map with selectable layers and a contextual analysis panel.

Map layer	Function
Site boundaries	Project and operable-unit delineation
Satellite imagery	Historical and current surface observations
Sampling locations	Field and laboratory records
Contaminant distribution	Measured and interpolated concentrations
Geochemistry	pH, moisture, redox, conductivity
Biological ecology	Organism observations and functional evidence
Climate	Historical and forecast conditions
Treatment areas	Proposed and active remediation zones
Monitoring	Results, progress and recovery indicators

A critical distinction: the application must visibly distinguish measured data, remotely sensed observations, interpolated surfaces and model predictions. Each should carry its own provenance and uncertainty.

3. The conversational intelligence layer
This is where the ChatGPT-like interface becomes particularly valuable.

Perhaps appropriately recognized as the DIGGIT Research and Remediation Agent.

It should function as an intelligent intermediary between the scientific datasets and the people making decisions.

DIGGIT Intelligence Engine
Proposed

User inquiry
"Which remediation approaches are most appropriate for this site's benzene and PAH distribution under projected seasonal moisture conditions?"

- Retrieve and aggregate
  Collect relevant site measurements, satellite observations, research records and climate data.
- Analyze and interpolate
  Apply geostatistical methods to spatially distributed observations, with uncertainty estimates and appropriate validation.
- Evaluate scientific evidence
  Compare candidate organisms, chemical processes and physical treatments against compound-specific evidence.
- Model and forecast
  Assess environmental conditions, treatment duration, resource requirements and potential scheduling conflicts.
- Present decision alternatives
  Explain the evidence, assumptions, costs, risks and unresolved questions behind each candidate protocol.

Decision-support output
A ranked set of candidate strategies, proposed monitoring intervals, estimated resource requirements and an auditable explanation of the recommendation.

One important architectural distinction: the language model should coordinate and explain the analysis, not independently invent spatial estimates. GIS interpolation, geochemical calculations and predictive models must be separate, testable computational services. The AI agent orchestrates them and communicates their results.

4. Remediation scheduling and optimization
This is the part that could make DIGGIT unusually useful.

Instead of merely recommending a treatment, it should help determine when, where, under what conditions and at what resource cost a treatment is most appropriate.

Optimization domain	Example
Financial	Treatment costs, transport, materials and labor
Environmental	Runoff, contaminant mobilization and ecological disturbance
Climate	Seasonal moisture, temperature and precipitation
Scheduling	Treatment windows, dependencies and monitoring intervals
Performance	Expected contaminant reduction and ecological recovery
Uncertainty	Missing measurements, model confidence and evidence strength

For example, a predicted period of heavy rainfall could trigger a review of a proposed soil-conditioning schedule because of runoff and contaminant-mobilization concerns. The platform would flag the conflict and suggest alternative scheduling scenarios, not automatically change an approved remediation plan.

5. The community research environment

The community component is essential. DIGGIT should allow researchers, environmental engineers, universities, remediation organizations and local participants to contribute to a shared knowledge base.

- Researcher profiles and attribution.
- Collaborative site projects.
- Shared analytical datasets and protocols.
- Version-controlled treatment proposals.
- Peer review and evidence grading.
- Field monitoring submissions.
- Project budgets and implementation records.
- Reproducible analysis histories.

Contributors should retain attribution, and access permissions should distinguish public data, restricted site information and unpublished research. In addition, articulating metadata with hyperlinks to precise and relevant research literature could be useful in sourcing analytics and deployment pool employment networks.

6. Proposed application architecture

A map based GUI interface aggregates a multilevel dataset interpolation model to provide personable performance data and compile optimization graphing analysis to

provide users with artifacts and datasets while integrating this research datum into systems' databases for community based, future implementation and developments.

DIGGIT Web Application
Interactive map · Research workspace · Project management · AI assistant

Geospatial engine
MapLibre · PostGIS

Data integration
Satellite and field APIs

Scientific engine
Python · SciPy · GeoPandas

Predictive engine
Climate and ML models

Shared research database
Sites · Samples · Measurements · Literature · Organisms · Protocols · Monitoring · Budgets

A practical implementation would use HTML/JavaScript for the mobile-accessible interface, MapLibre for mapping, PostgreSQL/PostGIS for spatial data, and Python/FastAPI for analytical services.

7. The first demonstrator: Del Amo Environmental Recovery Project
I would use Del Amo as the initial demonstration site, but keep the platform architecture independent of that location.

The first release should demonstrate five capabilities:

- Import and display site boundaries, available EPA records and satellite imagery.
- Register and map site-specific samples and analytical measurements.
- Display contaminant distributions, clearly distinguishing observations from interpolation.
- Compare scientific evidence for candidate benzene, ethylbenzene and PAH remediation approaches, including the separate Cra20 plant-support research hypothesis.
- Generate a climate-aware remediation planning report with costs, scheduling constraints, uncertainties and monitoring recommendations.

Site particulars introduces a unique opportunity for sedimentary remediation research applications and particulate containment protocols. It is also a prime location to garner insights into urban remediation practices and promote a science based approach to municipal health issues through social gardening techniques such as aesthetic based approaches to remediation therapeutics like the water reclamation and treatment services at Tongva Park in Santa Monica. Furthermore, pollutant particles present opportunities to develop fundamental technologies and techniques that could improve efficiency of existing biochemical processes and reduce timeframes for phytoextraction procedures and mitigate urban hydrocarbon pollution providing a practical framework for similar EPA Superfund National Priority sites.

Atmospheric Remediation Within Reach

The key deliverable is not a map with an AI chatbot attached. It is a reproducible environmental decision-support system with a conversational interface. The goal is to facilitate remediation processes while simultaneously furthering available resources for remediation solutions globally. By providing a community based, efficiency optimized dataset aggregate solution to ecological dysfunction, the Digital Interface for Graphing Geochemical Interpolation Treatment, DIGGIT, will be a fundamental tool for institutions, organizations, and landholders in developing remediation therapy solutions to geological contamination issues. With great care and concentrated effort, atmospheric remediation is within reach and environmental issues become geocivil solutions.

7 November, 2026

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